MATHS MANTHAN
• TUITION CLASSES •
e-mail: contact@mathsmanthan.co.in | Helpline: +91-8171546558

Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 02** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 9: Some Applications of Trigonometry** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 10(iii).
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer question (2 marks) and 3 Short Answer questions (3 marks each).
4. **Section C** comprises 1 Applied Case Study question of 4 marks with internal choice.
5. Use of calculators is not permitted. Take $\sqrt{3} = 1.732$ unless stated otherwise.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. The angle of elevation of the top of a tower at a point on the ground 50 m away from the foot of the tower is 45°. Then the height of the tower (in metres) is: [1]

(A) 25 m (B) 50 m (C) $50\sqrt{3}$ m (D) $\frac{50}{\sqrt{3}}$ m

Q2. The angle of depression of a car, standing on the ground, from the top of a 75 m high tower is 30°. The distance of the car from the base of the tower (in metres) is: [1]

(A) $25\sqrt{3}$ (B) $50\sqrt{3}$ (C) $75\sqrt{3}$ (D) 150

Q3. The angle of depression of a car parked on the road from the top of a 150 m high cliff is 30°. The distance of the car from the base of the cliff (in metres) is: [1]

(A) $50\sqrt{3}$ (B) $150\sqrt{3}$ (C) 150 (D) 75

Q4. A ladder 15 m long just reaches the top of a vertical wall. If the ladder makes an angle of 60° with the wall, then the height of the wall is: [1]

(A) $15\sqrt{3}$ m (B) $\frac{15\sqrt{3}}{2}$ m (C) 7.5 m (D) 15 m

Q5. **Assertion (A):** When the angle of elevation of the sun increases from 30° to 60°, the length of the shadow of a vertical tower decreases.
Reason (R): For a fixed height h, shadow length $s = h\cot\theta$, and $\cot\theta$ is a strictly decreasing function for $0° < \theta < 90°$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

SECTION B: Short Answer Questions (11 Marks)

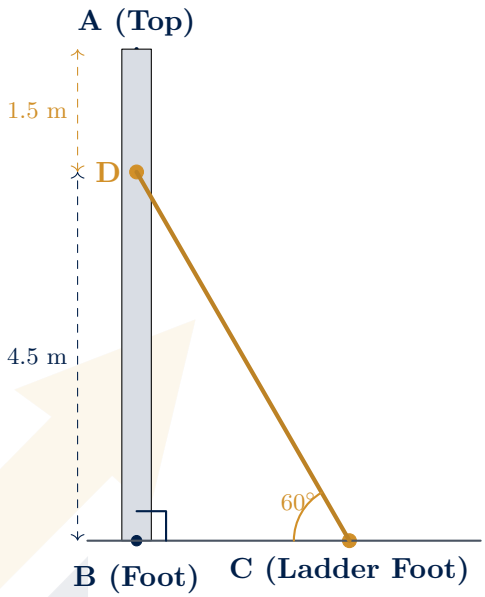
Q6. A ladder leaning against a vertical wall makes an angle of 60° with the horizontal ground. If the foot of the ladder is 9.5 m away from the wall, calculate the length of the ladder. [2]

Q7. A statue 1.6 m tall stands on the top of a pedestal. From a point on the ground, the angle of elevation of the top of the statue is 60° and from the same point the angle of elevation of the top of the pedestal is 45°. Find the height of the pedestal. [Take $\sqrt{3} = 1.732$] [3]

Q8. A TV tower stands vertically on a bank of a canal. From a point on the other bank directly opposite the tower, the angle of elevation of the top of the tower is 60°. From another point 20 m away from this point on the line joining this point to the foot of the tower, the angle of elevation of the top of the tower is 30°. Find the height of the tower and the width of the canal. [3]

Q9. The shadow of a vertical tower on level ground is found to be 40 m longer when the sun's altitude is 30° than when it is 60°. Find the height of the tower. [3]

Q10. CASE STUDY: Power Utility Infrastructure Maintenance



An electrical distribution technician has to perform emergency repair work on an overhead transmission pole AB of height 6 m. To maintain mandatory safety clearances from live high-voltage conductors at the top of the pole (A), safety protocols dictate that the technician must reach a working platform point D located exactly 1.5 m below the top.
A lightweight, heavy-duty insulated ladder CD is inclined at an angle of 60° to the horizontal ground to enable the technician to reach position D.

Based on the above mathematical scenario, answer the following questions:

(i) What is the vertical height from the ground to the repair position point D? [1]

(ii) What should be the minimum length of the ladder CD required to reach point D? [Take $\sqrt{3} = 1.732$] [1]

(iii) How far from the foot of the pole (B) should the foot of the ladder (C) be placed on the ground? [2]

— OR —

If ground obstructions require the ladder to be placed at an inclination of 45° instead of 60° to reach the same height of 4.5 m, find the new required length of the ladder. [2]

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★